

Sanvi Kaushik

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TECHNICAL SKILLS

Languages: Python, C++, JavaScript, C, SQL, Java, Bash

Frameworks & Tools: React.js, Node.js, REST/gRPC, Docker, GitLab CI/CD, Jenkins

Libraries & Systems: Linux, PyTorch, TensorFlow Lite, scikit-learn, NumPy, Pandas, PostgreSQL, Redis, MongoDB

EDUCATION

Carleton University

Bachelor of Engineering (BEng) in Software Engineering, Co-operative Education

Ottawa, ON

Expected Apr. 2028

- **Awards:** Dean's Honour List, President's Scholarship (\$12,000)

GPA: 3.73/4.0

EXPERIENCE

Software Developer Intern

Ericsson

May 2025 – Present

Ottawa, ON

- Built a **retrieval-augmented chatbot** for **Cloud RAN** documentation, benchmarked across LLMs, and deployed to an internal tool used by **20+ developers**
- Implemented **C++ protocol encoding** for beamforming modules in **Linux**, passing **200+ CI Jenkins pipelines** to support a major **5G virtualization upgrade**
- Created **20+ Google Tests** validating message encode/decode, increasing unit-test coverage to **100%**

Software Developer Intern

BlackBerry

May 2024 – Aug. 2024

Ottawa, ON

- Engineered a **C++ synthetic sensor** in the **BlackBerry IVY[®]** cloud to process vehicle telemetry, improving dog-safety detection accuracy by **40%**
- Trained and deployed a **TensorFlow Lite** model for seat-occupancy detection with **92% precision**, integrating inference into the Android application using **Kotlin**
- Refactored **GitLab CI/CD** pipelines using **YAML** and **Docker** in a **Linux** environment, stabilizing **100+ builds** with **30% fewer failures**

PROJECTS

Learning-Driven Cache Eviction System | *Python, PyTorch, XGBoost, NumPy, scikit-learn, Docker, Redis*

- Built a **HashMap-doubly linked list** cache with an **RL-XGBoost eviction** policy, raising hit rate from **72% → 88%**
- Used **XGBoost + $O(\log \text{cache_size})$** scoring to cut **99th-percentile miss latency by 34%**
- Achieved **$O(1)$ RL inference** with a **<10K-parameter PyTorch DQN**, improving eviction decision time by **2.3×**

Trace (1st Place @ Tech-Nol-Hack 2025) | *JavaScript, Node.js, Express, SQLite, Leaflet, Render*

- Developed a **risk-aware routing engine** computing detours in **<200 ms**, raising routing safety scoring by **32%**
- Designed a **privacy-preserving reporter** using anonymous clustering, improving data reliability by **45%**
- Awarded **1st Place** and \$450 prize by a **Microsoft Security Engineer** for measurable civic impact

Adaptive LeetCode Skill Agent | *Python, NumPy, scikit-learn, XGBoost, FastAPI, SQL, JavaScript*

- Modeled user skill with **Bayesian logistic regression** and **boosted trees**, raising targeted-question recall by **31–38%**
- Optimized question selection via a **contextual bandit** (UCB + posterior sampling), reducing substep errors by **34%**
- Built a **Chrome extension + FastAPI** pipeline enabling online updates over **200+ episodes** with sub-**5 ms** inference